

Sample Space :

$$\text{Toss a coin} \rightarrow \Omega = \{HH, HT, TH, TT\}$$
$$P \rightarrow p_1, p_2, p_3, p_4$$

$$\sum p_i = 1$$

$$p_i \geq 0$$

event

$$A = \{ \text{At least 1 T appears} \}$$
$$= \{ HT, TH, TT \}$$

$$P(A) = p_2 + p_3 + p_4 = 1 - p_1$$
$$= 1 - P(A')$$

$$\mathcal{F} = \{ \text{subsets of } \Omega \} \rightarrow \underline{\text{ } \sigma\text{-field}}$$

A random variable X is a function $f: \Omega \rightarrow (-\infty, \infty)$

$X = \# \text{ of heads}$

$$X(\omega_1) = 2$$

$$X(\omega_2) = 1$$

$$X(\omega_3) = 1$$

$$X(\omega_4) = 0$$

$$X_2 = 5X_1$$

$$X_3 = \#H'_A - \#T'_A$$

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$X \sim \text{Bin}(n, p)$, $X = \text{H's in } n \text{ tosses}$ where $p = P(H)$

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } k=0, \dots, n$$

$$X \sim \text{Bernoulli}(p) \Leftrightarrow X \sim \text{Binomial}(1, p)$$

$Y = \text{longest run of H's}$

$X_1, X_2, \dots$ on $\Omega = \{X_i : i \in \mathbb{N}\} \rightarrow \text{discrete}$

$\{X_t : t \geq 0\} \rightarrow \text{continuous}$
Stochastic Process $\rightarrow$ Sequence of these random variables

$F(x) = P(X \leq x) \rightarrow \text{Cumulative Distribution Function CDF}$

X_1, X_2 are independent if $P(X_1 \leq x_1, X_2 \leq x_2) = P(X_1 \leq x_1) \cdot P(X_2 \leq x_2)$

$X_1, X_2, \dots, X_n$ are mutually independent if

$$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = P(X_1 \leq x_1) \cdot P(X_2 \leq x_2) \dots P(X_n \leq x_n)$$

for all $x_1, x_2, \dots, x_n \in (-\infty, \infty)$

(then ~~also~~ conversely true)

If X_i and X_j are independent for all i and j

Throw a fair dice twice.

$X_1 =$ Sum of the outcomes is 7

$X_2 =$ ~~the first throw of the dice in~~

X_1 = Sum of the outcomes is 7
 X_2 = ~~the first throw of the dice is~~ The outcome of first throw
 X_3 = ~~the first throw of the dice is~~ " " " second "

$$\Omega = \{ (i, j) : 1 \leq i, j \leq 6 \}$$

$$P(X_2 = \overset{L}{6}, X_3 = \overset{L}{6}) = P(X_2 = \overset{L}{6}) \cdot P(X_3 = \overset{L}{6})$$

X_1 and X_2 are independent

~~XXXXXXXXXXXX~~

$$P(X_1 = j, X_2 = k) = \begin{cases} 0 & \text{otherwise} \\ \frac{1}{36} & X_3 = j - k \end{cases}$$

$$P(X_1=7, X_2=5, X_3=4) = \frac{1}{36}$$

$$\neq \underbrace{P(X_1=7)}_{\frac{1}{6}} \cdot \underbrace{P(X_2=3) \cdot P(X_3=4)}_{\frac{1}{36}}$$

Defn $X_1, X_2, \dots$ is said to be independent sequence of r.v's if $P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$ for all $-\infty \leq x_1, x_2, \dots, x_n \leq \infty$ and for all $n \geq 1$

Defn. $X_1, X_2, \dots$ is said to be an independent and identically distributed sequence (iid) of r.v's if

(i) $X_1, X_2, \dots$ is an indep seq. of r.v's

(i) $X_1, X_2, \dots$ is an indep. seq. of r.v.
(ii) for all i, j $P(X_i \leq x) = P(X_j \leq x)$ for all $x \in \mathbb{R}$

HW: X, Y, Z mutually independent r.v's with ~~also~~ also identically distributed with $P(X=1) = \frac{1}{2}$, $P(X=-1) = \frac{1}{2}$. Consider the r.v's XY, YZ, XZ .

Q. Are they ^{mutually} independent? _/_/_

For a discrete r.v X : $E(X) = \sum x_i P(X=x_i)$

where $x_1, x_2, \dots$ are the values taken by X

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2$$

$$\begin{aligned} \text{For r.v's } X, Y: \text{Cov}(X, Y) &= E((X - E(X))(Y - E(Y))) \\ &= E(XY) - E(Y)E(X) \end{aligned}$$

HW: 2) $X_1, X_2, \dots, X_{n-1}$ iid Bernoulli (p)

$$X_n = \begin{cases} 1 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is odd} \\ 0 & \text{if } \sum_{i=1}^{n-1} X_i \text{ is even} \end{cases}$$

Q. Is the sequence $X_1, \dots, X_n$ indep? ~~Yes~~

Is the sequence $X_2, \dots, X_n$ indep? ~~Yes~~

Is the sequence $X_1, X_3, \dots, X_n$ indep?

$\{a_0, a_1, \dots\}$ finite or infinite

The generating function of this sequence is defined as

$$A(s) = \sum_{i=0}^{\infty} a_i s^i = \lim_{n \rightarrow \infty} \underbrace{\sum_{i=0}^n a_i s^i}_{\text{polynomial of } s \text{ in degree } n}$$

for $|s| < R$ where

R is Radius of Convergence

$$\{1, 1, 1, \dots\} \rightarrow A(x) = \sum_{i=0}^{\infty} 1 \cdot x^i$$

$$= 1 + x + x^2 + x^3 + \dots$$

$$= \frac{1}{1-x} \quad \text{for } |x| < 1$$

↓
Radius of convergence

X is a non negative integer valued r.v

$$X = k, \quad k \geq 0$$

$$P(X=k) = p_k \Rightarrow \left(\sum_{k=0}^{\infty} p_k = 1 \right)$$



$$P(s) = \sum_{k=0}^{\infty} p_k s^k \rightarrow \text{Probability Generating Function}$$

Example: 1) $X \sim \text{Bernoulli}(p)$

$$X = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$P_X(s) = (1-p)s^0 + ps^1$$

2) $X \sim \text{Binomial}(n, p)$

$$P[X=k] = {}^nC_k p^k (1-p)^{n-k} \text{ for } k \in [0, n]$$

$$P_X(s) = \sum_{k=0}^n {}^nC_k p^k (1-p)^{n-k} s^k = \sum_{k=0}^n {}^nC_k (ps)^k (1-p)^{n-k}$$

$$\text{Binomial Theorem: } (a+b)^n = \sum_{k=0}^n {}^nC_k a^k b^{n-k}$$

$$\Rightarrow P_X(s) = (ps + 1-p)^n$$

→ Same as Bernoulli with $n=1$

3) $X \sim \text{Geometric}(p)$

X : Number of Failures till first Success
 $X \in [0, \infty)$

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$$P(X=k) = (1-p)^k p \text{ for } k=0, 1, \dots$$

$$P_X(s) = \sum_{k=0}^{\infty} (1-p)^k p s^k = p \left(\sum_{k=0}^{\infty} ((1-p)s)^k \right)$$

$$= \frac{p}{1-s(1-p)}$$

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In general: $P_X(s) = \sum_{k=0}^{\infty} p_k s^k$

 for all $|s| \leq 1$

→ how!

$$P_X(1) = 1$$

6) $X \sim \text{Poisson}(\lambda)$

$$P[X=k] = \frac{e^{-\lambda} \lambda^k}{k!} \text{ for } k \geq 0$$

$$e^x = \underbrace{x^0}_{1!} + \underbrace{x^1}_{1!} + \underbrace{x^2}_{2!} + \underbrace{x^3}_{3!} + \dots$$

$$P_X(s) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} s^k = \sum_{k=0}^{\infty} \frac{e^{-\lambda} (\lambda s)^k}{k!}$$

$$= (e^{-\lambda}) (e^{\lambda s})$$

$$P_X(s) = e^{-\lambda(1-s)}$$

$$E(s^X) = \sum_{k=0}^{\infty} P[X=k] s^k$$

for a r.v X non neg integer valued

$$P_X(s) = \sum_{k=0}^{\infty} p_k s^k \quad |s| \leq 1$$

$$= p_0 s^0 + p_1 s^1 + p_2 s^2 + \dots$$

$$\frac{dP_X(s)}{ds} = 0 + p_1 + 2p_2 s + 3p_3 s^2 + \dots$$

$$\left. \frac{d}{ds} P_X(s) \right|_{s=0} = \mu_1 = P[X=1]$$

$$\left. \frac{d^2}{ds^2} P_X(s) \right|_{s=0} = \mu_2 = [2 P[X=2]]$$

$$\boxed{\left. \frac{d^n}{ds^n} P_X(s) \right|_{s=0} = [n P[X=n]]}$$

$$a = \{a_0, a_1, a_2, \dots\} \rightarrow A(s)$$

$$b = \{b_0, b_1, b_2, \dots\} \rightarrow B(s)$$

$$C_n = a_0 b_n + a_1 b_{n-1} + \dots + a_{n-1} b_1 + a_n b_0$$

$$= \sum_{i=0}^n a_i b_{n-i}$$

$$C = \{C_0, C_1, \dots\} \rightarrow C(s)$$

C is convolution of a and b

$$A(s)B(s) = (a_0 + a_1 s + a_2 s^2 + \dots)$$

$$\times (b_0 + b_1 s + b_2 s^2 + \dots)$$

$$= a_0 b_0 + s(a_1 b_0 + a_0 b_1) + s^2(a_2 b_0 + a_1 b_1 + a_0 b_2) + \dots$$

$\nearrow C_1 \qquad \qquad \qquad \nearrow C_2$

$$\Rightarrow A(s)B(s) = C(s)$$

Independent $\left\{ \begin{array}{l} X \text{ taking values } 0, 1, 2, \dots \\ Y \text{ taking values } 0, 1, 2, \dots \end{array} \right.$

$$\Rightarrow P[X=k, Y=L] = P[X=k] P[Y=L] \text{ for all } k, L$$

$$\begin{aligned}
 P(X+Y=j) &= P[X=0, Y=j] + P[X=1, Y=j-1] + \dots + P[X=j, Y=0] \\
 &= P(X=0)P(Y=j) + P(X=1)P(Y=j-1) + \dots + P(X=j)P(Y=0) \\
 &= p_0^x p_j^y + p_1^x p_{j-1}^y + \dots + p_j^x p_0^y
 \end{aligned}$$

$$P_{X+Y}(s) = P_X(s)P_Y(s) \text{ for } X, Y \text{ indep. r.v.s}$$

Notation: if the seq C is a convolution of the sequences a and b , then we write $C = a * b$

$$P_W(s) = P_Z(s) \iff W \text{ and } Z \text{ have same probability distribution, i.e. } W \text{ is same r.v. as } Z$$

(Proof by taking derivatives, and obtaining same probability at all points by setting $s=0$)

$$X_1, X_2, X_3, \dots, X_n \text{ iid Bernoulli}(p)$$

$$P_{X_k}(s) = (1-p) + ps = P_{X_k}(s), \quad k \in [1, n]$$

$$\begin{aligned}
 P_{X_1+X_2+X_3+\dots+X_n}(s) &= (1-p+ps)^n \\
 &= P_{\text{Binomial}(n,p)}(s)
 \end{aligned}$$

$$\Rightarrow (X_1 + X_2 + X_3 + \dots + X_n) \sim \text{Binomial}(n, p)$$

$$\left. \begin{array}{l} X_1 \sim \text{Poisson}(\lambda) \\ X_2 \sim \text{Poisson}(\mu) \end{array} \right\} \text{ indep}$$

$$\begin{aligned} P_{X_1+X_2}(s) &= P_{X_1}(s) \cdot P_{X_2}(s) \\ &= e^{-\lambda(1-s)} \cdot e^{-\mu(1-s)} = e^{-(\lambda+\mu)(1-s)} \\ &= P_{\text{Poisson}(\lambda+\mu)} \end{aligned}$$

$$\therefore (X_1 + X_2) \sim \text{Poisson}(\lambda + \mu)$$

$$E(X^k) = m_k \rightarrow k^{\text{th}} \text{ moment}$$

$$\frac{d^k}{ds^k} P_X(s) = \sum_{i=0}^{\infty} i(i-1)\dots(i-k+1) p_i s^{i-k}$$

$$\left. \frac{d}{ds} P_X(s) \right|_{s=1} = \sum_{i=0}^{\infty} i p_i = E(X)$$

$$\left. \frac{d^2}{ds^2} P_X(s) \right|_{s=1} = \sum_{i=1}^{\infty} i(i-1) p_i = E[X(X-1)]$$

$$\left. \frac{d^k}{ds^k} P_X(s) \right|_{s=1} = \underbrace{E[X(X-1)(X-2)\dots(X-k+1)]}_{k^{\text{th}} \text{ factorial moment of } X}$$

$$P_X'(1) = E[X]$$

$$P_X''(1) = E[X(X-1)] = E[X^2 - X] = E[X^2] - E[X]$$

$$\Rightarrow E[X^2] = P_X''(1) + P_X'(1)$$

If X and Y are 2 rv's $P(X=k|Y=L)$
 $= \frac{P(X=k, Y=L)}{P(Y=L)}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} \text{Var}[X] &= E[X^2] - (E[X])^2 \\ &= P_X''(1) - (P_X'(1))^2 \end{aligned}$$

drunk

A person takes a step in unit time

$$X_k = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

X_k : Action at $t=k$

0 $\rightarrow$ stays, 1 $\rightarrow$ moves forward
 at time $= N$, he collapses
 No indep of steps

$$S_N = \sum_{i=1}^N X_i$$

how far he
has moved

Theorem: Let $X_1, X_2, \dots$ be iid rv's and N a rv independent of $X_1, X_2, \dots$. Let $S_N = \sum_{i=1}^N X_i$

then $P_{S_N}(\lambda) = P_N(P_{X_1}(\lambda))$

Proof: $P_{S_N}(\lambda) = \sum_{i=0}^{\infty} P(S_N=i) \lambda^i$

$$= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P(S_N=i | N=j) P(N=j)$$

$$= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P(S_N=i, N=j)$$

$$\sum_{j=0}^{\infty} P(S_N=i, N=j) = P(S_N=i)$$

$$= \sum_{i=0}^{\infty} \lambda^i \sum_{j=0}^{\infty} P\left(\sum_{l=1}^j X_l = i \mid N=j\right)$$

$$= \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \lambda^i P(N=j) P\left(\sum_{l=1}^j X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) \sum_{i=0}^{\infty} s^i P\left(\sum_{l=0}^j X_l = i\right)$$

$$= \sum_{j=0}^{\infty} P(N=j) (P_{X_1}(s))^j$$

$\searrow$
 t

$$= \sum_{j=0}^{\infty} P(N=j) t^j = P_N(t)$$

$$= P_N(P_X(s))$$

Example:

$X_1, X_2, \dots$ i.i.d.

$$X_1 = \begin{cases} 0 & \text{w.p. } \frac{1}{2} \\ 1 & \text{w.p. } \frac{1}{2} \end{cases}$$

$$P_{X_1}(s) = \frac{1}{2} + \frac{1}{2}s$$

$N \sim \text{geometric}\left(\frac{2}{3}\right)$

$$P_N(s) = \frac{2}{3-s}$$

$$P_{SN}(s) = P_N(P_{X_1}(s))$$

$$= P_N\left(\frac{1+s}{2}\right) = \frac{2}{3 - \left(\frac{1+s}{2}\right)} = \frac{4}{5-s}$$

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$$\text{pgf} : P_X(s) = E(s^X)$$

$$\text{mgf} : M_X(s) = E(e^{sx}) = \sum_{i=0}^{\infty} e^{si} P[X=i]$$

$$\left. \frac{d}{ds} M_X(s) \right|_{s=0} = \sum_{i=0}^{\infty} i P[X=i] = E[X]$$

$$\left. \frac{d^2}{ds^2} M_X(s) \right|_{s=0} = \sum_{i=0}^{\infty} i^2 P[X=i] = E[X^2]$$

$$\left. \frac{d^k}{ds^k} M_X(s) \right|_{s=0} = E[X^k]$$

BRANCHING PROCESS
galton-watson process

of children of an individual is

0	w.p. $\frac{1}{2}$
1	w.p. $\frac{1}{4}$
2	w.p. $\frac{1}{4}$

gen 0 : 1 individual

